

Project Nimbus — Internal Overview

Project Nimbus is an internal weather-analytics platform built by Artilence. It was launched on March 14, 2023, and is led by Dr. Elena Ruiz, the company's Chief Data Officer. The initial version was delivered by a core team of seven engineers over an eight-month build.

The platform is written primarily in Python, with the performance-critical ingestion path implemented in Rust. Nimbus runs on a private Kubernetes cluster and is not exposed to the public internet.

Mission

The mission of Project Nimbus is to give operations teams earlier and more accurate severe-weather warnings, reducing costly false alarms without missing genuine events.

Project Nimbus — Modules and Metrics

Nimbus is organised into three core modules. Skywatch produces the forecasts. Cirrus handles data ingestion from field sensors. Stratus is the alerting engine that notifies on-call staff.

In production, Nimbus processes about 2.3 million sensor readings per day. Since launch it has reduced the false-alarm rate from 18 percent to 4 percent. Raw sensor data is kept under a 90-day retention policy, after which it is aggregated and purged.

Roadmap

The next major milestone is multi-region support, scheduled for the fourth quarter of 2024. A public read-only API is under discussion but has not been approved.